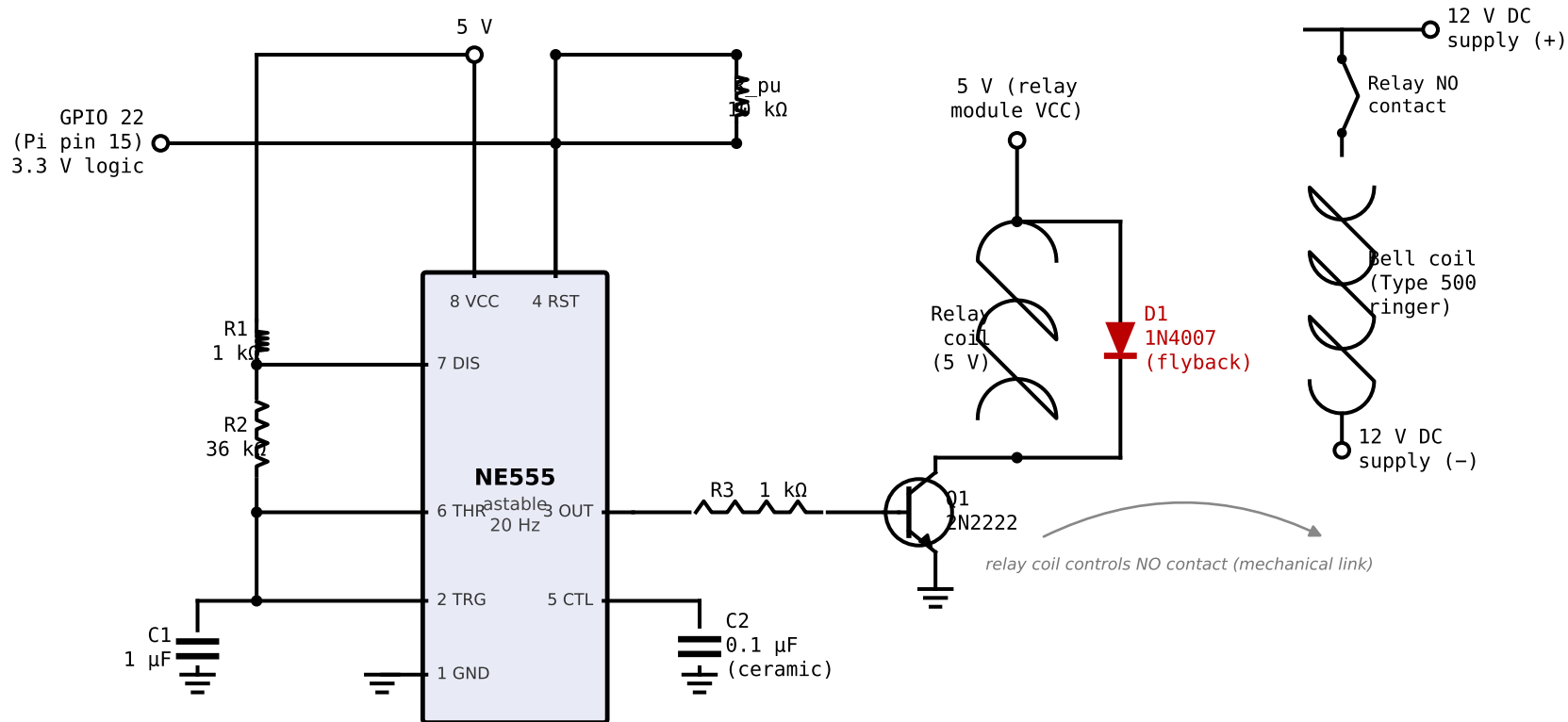


Bell Wiring — Option 2: 555 Astable at 20 Hz, Gated by GPIO 22

Authentic mechanical ring — clapper oscillates at 20 Hz during ring cadence



555 astable frequency: $f = 1.44 \div ((R1 + 2 \times R2) \times C1) = 1.44 \div (73\,000 \times 0.000001) \approx 19.7 \text{ Hz}$ (nominal 20 Hz)

GPIO 22 HIGH $\rightarrow$ RESET (pin 4) released $\rightarrow$ 555 oscillates $\rightarrow$ 20 Hz square wave on OUT $\rightarrow$ Q1 switches relay at 20 Hz $\rightarrow$ clapper oscillates $\rightarrow$ authentic bell ring.

GPIO 22 LOW $\rightarrow$ RESET held LOW $\rightarrow$ 555 output frozen LOW $\rightarrow$ relay open $\rightarrow$ bell silent. Ring cadence (2 s on / 4 s off) is still controlled by the script.

GPIO 22 is 3.3 V logic; 555 RESET LOW threshold $\approx 0.7 \text{ V}$ for 5 V supply – direct connection works, no level-shifter needed.

R_{pu} (10 k Ω): pull-up from 5 V to RESET – keeps RESET HIGH if GPIO floats at boot, preventing spurious ringing during startup.

Q1 (2N2222 NPN): R3 = 1 k Ω base resistor. Handles relay coil current $\leq 100 \text{ mA}$. Substitute 2N3904 for coils $\leq 50 \text{ mA}$.

D1 (red) – flyback diode: ESSENTIAL across relay coil; cathode (banded end) toward 5 V rail. Without it Q1 fails instantly on de-energise.

C2 (0.1 μF ceramic): bypass cap on 555 pin 5 (CTL) – suppresses VCC noise coupling into timing; place as close to IC as possible.

Tune frequency: R2 = 27 k Ω $\rightarrow$ $\approx 25 \text{ Hz}$; R2 = 47 k Ω $\rightarrow$ $\approx 15 \text{ Hz}$. Bell electromagnet is rated 20 Hz; stay within 16–25 Hz for reliable ringing.